F21RO Intelligent Robotics

Coursework 1 - Inverse Kinematics Neural Network

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1. Introduction

Inverse Kinematics (IK) estimates robot joint configurations needed to reach a desired TCP position, but can be challenging for different geometries and produce multiple solutions

Aim: Train and evaluate a single Artificial Neural Network capable of approximating IK for both Ned and SCARA.

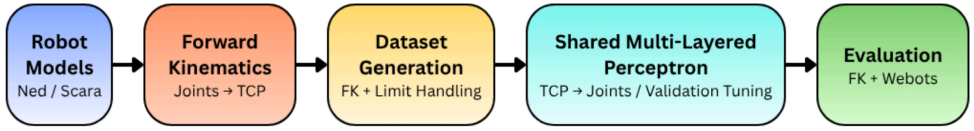


Fig.1 Niryo Ned



Fig. 2 Epson Scara

2. Method Overview



*Webots uses TCP positions relative to ARM coordinate frame.

Robot	Active Joints	Fixed / Ignored
Ned	q1, q2, q3 (revolute)	Remaining joints fixed
SCARA	q1, q2 (revolute), q4 (prismatic)	Shaft rotation fixed

Table.1 Robot Active Joints

3. Dataset & Network Design

- Joint samples within permitted ranges
- **Elbow-up restriction to reduce multiple-IK-solution ambiguity**
- FK(joints sample) → TCP position
- Datasets nested across sizes

Small ⊂ Medium ⊂ Large

- Independent test set of 500 samples per robot used for evaluation.
- Size count chosen to systematically evaluate how size affects accuracy and generalization while balancing time and cost

*Counts per Robot

Dataset	Size	Training:Validation (80:20 Split)
Small	1000	800 : 200
Medium	5000	4000 : 1000
Large	10000	8000 : 2000

Table.2 Datasets Sizes & Splitting

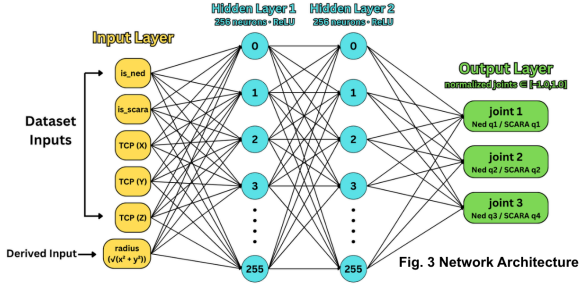


Fig. 3 Network Architecture

Input	Type	Meaning
is_ned	boolean (0 or 1)	1 if Ned, otherwise 0
is_scara	boolean (0 or 1)	1 if SCARA, otherwise 0
x	float	TCP X-coordinate
y	float	TCP Y-coordinate
z	float	TCP Z-coordinate
radius	float	Horizontal Radial Distance $\sqrt{(x^2 + y^2)}$

Table. 3 Input Format

Output	Type	Meaning
joint 1	float	q1 normalized in [-1,1]
joint 2	float	q2 normalized in [-1,1]
joint 3	float	q3 for Ned, q4 for SCARA normalized in [-1,1]

Table. 4 Output Format

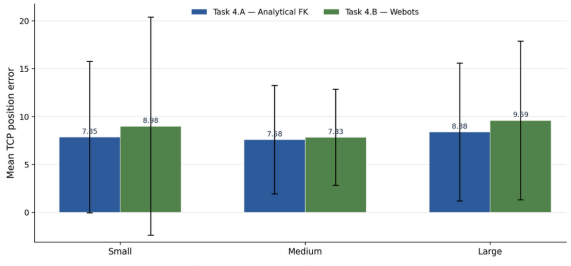
Hyper-parameter	Value
Layers	2
Neurons per Layer	256
Activation Function	ReLU
Optimizer	Adam
Learning Rate	0.001
Max Epochs	2000
Loss Function	Squared Error

Table. 5 List of Hyper-parameters

4. Results & Comparison

Layers	Activation	Learning Rate	Optimizer	Normalized Inputs	Radius Parameter	Clipping
64, 64	ReLU	0.001	Adam	True	False	True
128, 128	ReLU	0.0005	Adam	True	False	True
128, 128	Tanh	0.001	Adam	False	False	True
256, 256	**ReLU**	**0.001**	**Adam**	**False**	**True**	**True**

Table. 6 Validation Tuning



Graph. 1 Ned - TCP Error



Graph. 2 SCARA - TCP Error

Robot	Model	Task4.A - Anatical FK	Task4.B - Webots
NED	Small	7.846 ± 7.897	8.982 ± 11.370
NED	Medium	7.567 ± 5.66	7.830 ± 5.009
NED	Large	8.380 ± 7.194	9.588 ± 8.272
SCARA	Small	4.551 ± 2.793	4.721 ± 3.283
SCARA	Medium	4.750 ± 2.314	4.610 ± 2.296
SCARA	Large	2.582 ± 1.595	2.766 ± 1.823

Table. 7 TCP Position Error Summary

Robot	Model	Joint 1 (rad) Mean ± Std	Joint 2 (rad) Mean ± Std	Joint 3 Mean ± Std
Ned	Small	−0.02818 ± 0.13890	0.01302 ± 0.01254	0.00536 ± 0.00759 rad
Ned	Medium	−0.01919 ± 0.03844	0.01489 ± 0.01004	0.00360 ± 0.00484 rad
Ned	Large	0.02624 ± 0.04913	0.01074 ± 0.00804	0.00058 ± 0.00509 rad
SCARA	Small	−0.01090 ± 0.01865	0.02646 ± 0.04344	$6.8 \times 10^{-7} \pm 5.5 \times 10^{-6}$ m²
SCARA	Medium	−0.01804 ± 0.01366	0.02196 ± 0.02738	$1.6 \times 10^{-7} \pm 4.3 \times 10^{-7}$ m²
SCARA	Large	−0.00826 ± 0.01046	0.00967 ± 0.02232	$1.6 \times 10^{-7} \pm 1.25 \times 10^{-6}$ m²

Table. 8 Joint Errors of Ned & SCARA

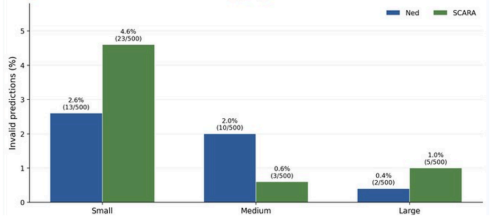


Table. 9 Joint-limit Violations Before Clipping

5. Discussion

- Larger datasets improved SCARA's accuracy, but Ned performed best with medium.
- Increased complexity did not guarantee better performance; adding radial distance significantly improved accuracy.
- Optimizing for one robot reduced performance for the other.
- IK Ambiguity: Multiple joint configurations for same TCPs complicates regression.
- Occasional joint-limit violations and sensitivity to out-of-distribution targets.

6. Conclusions

- Successfully trained one shared MLP to approximate IK for both Ned and SCARA.
- Demonstrated the feasibility of a shared NN across different kinematic structures.
- Future improvements include robot-specific networks, workspace validation and feedback-based correction.

References

Robots Image Reference: Coursework Specification